

The following is a summary of specific changes that have been made in the third edition:

- Chapter 1 now features the text editor VS Code, which is popular among beginner and professional programmers and works well on all operating systems.
- Chapter 2 includes the new methods `removeprefix()` and `removesuffix()`, which are helpful when working with files and URLs. This chapter also features Python's newly improved error messages, which provide much more specific information to help you troubleshoot your code when something goes wrong.
- Chapter 10 uses the `pathlib` module for working with files. This is a much simpler approach to reading from and writing to files.
- Chapter 11 uses `pytest` to write automated tests for the code you write. The `pytest` library has become the industry standard tool for writing tests in Python. It's friendly enough to use for your first tests, and if you pursue a career as a Python programmer, you'll use it in professional settings as well.
- The *Alien Invasion* project in Chapters 12–14 includes a setting to control the frame rate, which makes the game run more consistently across different operating systems. A simpler approach is used to build the fleet of aliens, and the overall organization of the project has been cleaned up as well.
- The data visualization projects in Chapters 15–17 use the most recent features of `Matplotlib` and `Plotly`. The `Matplotlib` visualizations feature updated style settings. The random walk project has a small improvement that increases the accuracy of the plots, which means you'll see a wider variety of patterns emerge each time you generate a new walk. All the projects featuring `Plotly` now use the `Plotly Express` module, which lets you generate your initial visualizations with just a few lines of code. You can easily explore a variety of visualizations before committing to one kind of plot, and then focus on refining individual elements of that plot.
- The Learning Log project in Chapters 18–20 is built using the latest version of `Django` and styled using the latest version of `Bootstrap`. Some parts of the project have been renamed to make it easier to follow the overall organization of the project. The project is now deployed to `Platform.sh`, a modern hosting service for `Django` projects. The deployment process is controlled by `YAML` configuration files, which give you a great deal of control over how your project is deployed. This approach is consistent with how professional programmers deploy modern `Django` projects.
- Appendix A has been fully updated to recommend current best practices for installing Python on all major operating systems. Appendix B includes detailed instructions for setting up VS Code, and brief descriptions of most of the major text editors and IDEs in current use. Appendix C directs readers to several of the most popular online resources for getting